

HW1 Virtuoso Repo: <https://github.com/vishalkevat007/Advanced-VLSI-Design-ECE695-Spring2026-Homeworks.git>

(a) Implement the following function using CMOS gates in cadence/HSPICE. The CMOS gates in your implementation should not have more than 3 stacked transistors. The complements of the primary inputs are not available i.e. if you need them, you will need to generate them. You may require more than one CMOS gate to implement the function.

$$F = ABC + A'B'C + AB'C' + A'BC' + CD + AD'$$

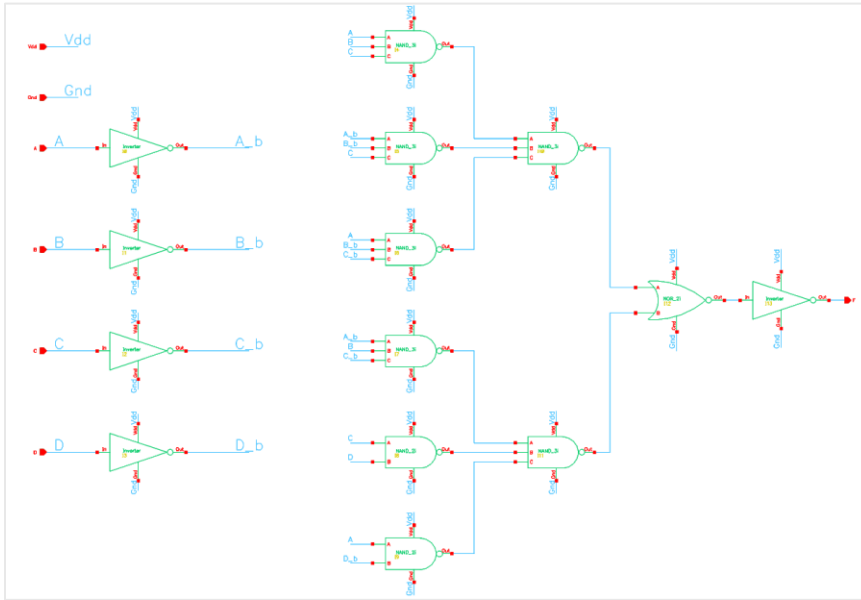


Fig 1: Cadence Schematic for F

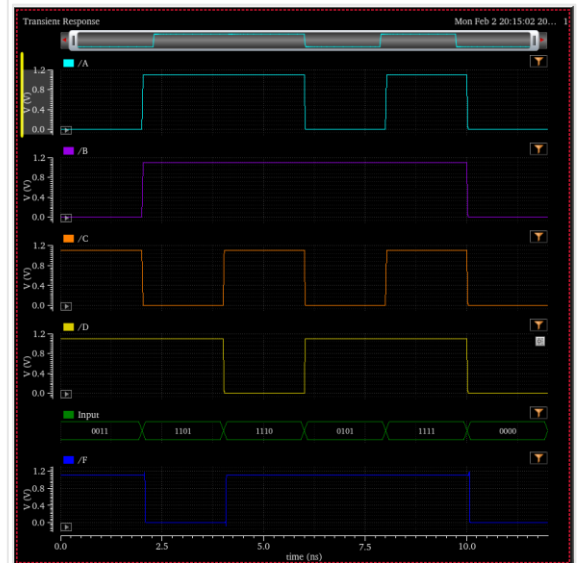


Fig 2: Functional Verification

Above figure shows implementation of function F with CMOS gates. The functional verification in Fig. 2 shows that the implementation of F using the gates is correct.

(b) Size the transistors to balance the pull-up and pull-down networks and obtain the leakage when all inputs are tied to 0 and when all inputs are tied to VDD.

All CMOS Gates seen in Fig. 1 are sized with equal pull-up and pull-down strength w.r.t reference inverter which has $W_n = 120n$ and $W_p = 180n$.

Leakage Current when all inputs are tied to '0' = 455.8 pA

Leakage Current when all inputs are tied to V_{DD} = 243.9 pA

Note: $V_{DD} = 1.1$ V

(c) Now use Shannon-based expansion and dynamic supply gating to implement the above function with A as the control variable. Your implementation should not have more than 3 stacked transistors, *including the gating transistor*. Pay particular attention to the sharing of the circuits amongst different co-factors.

$$F_{A=1} = (B \oplus C)' + CD + D'$$

$$F_{A=0} = B \oplus C + CD$$

Here, we can share the $B \oplus C$ and CD between the two co-factors. The final schematic looks like below Fig 3.

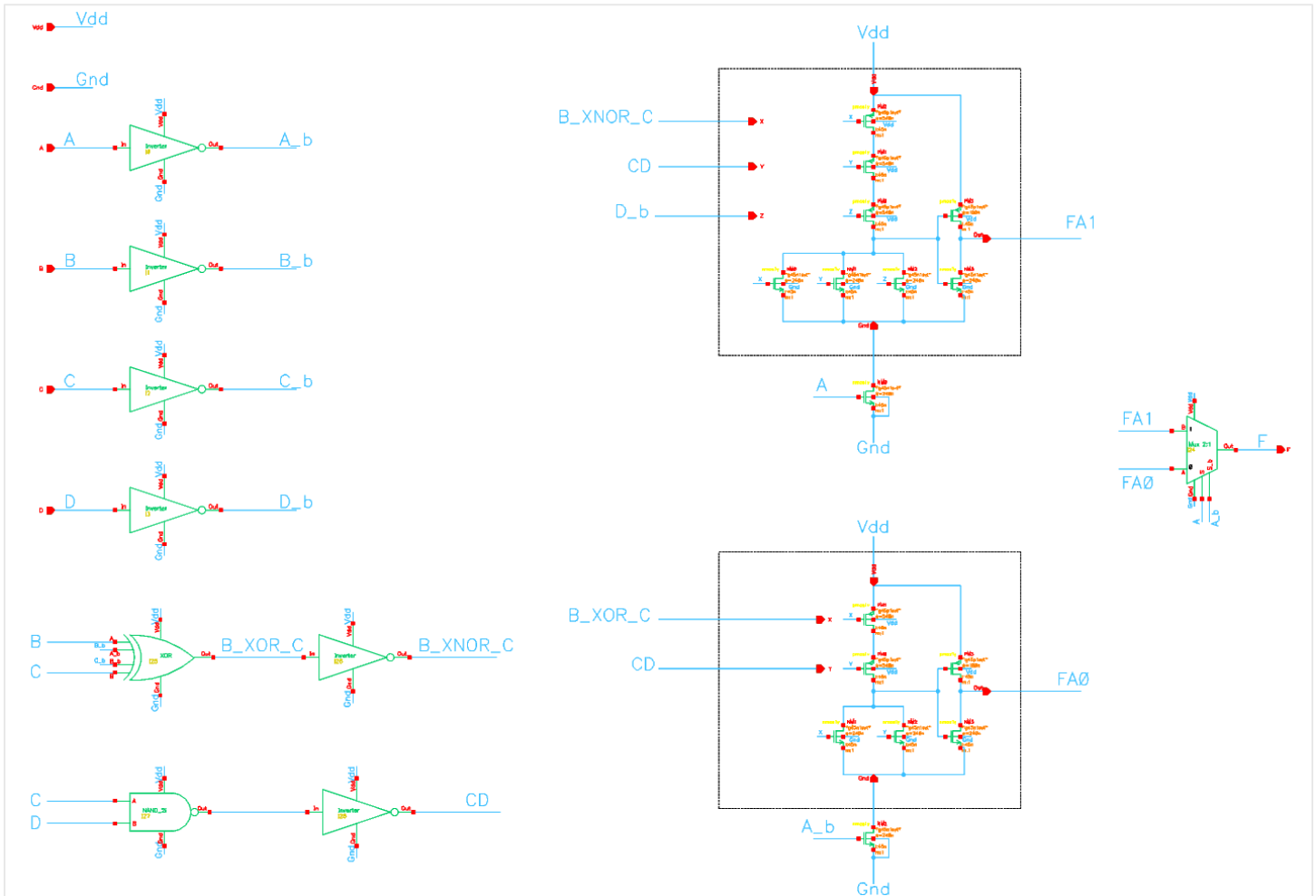


Fig 3: Function F with dynamic supply gating using variable A

(d) Size the transistors to balance the pull-up and pull-down networks and obtain the leakage when all inputs are tied to 0 and when all inputs are tied to VDD.

Firstly, Inverter was sized by keeping the width of NMOS $W_n = 120n$ and PMOS width W_p was swept to get equal rise and fall delays. We got $W_p = 180n$. Using this reference inverter, all other gates were sized using resistance equalization method.

Now, the pull-up and pull-down network of cofactor implementation (including the gating NMOS transistor) was sized according to resistance equalization to make the effective width of pull-up as $180n$ and pull-down as $120n$. This can be verified from Fig 4. which shows a 3 input OR gate for implementing FA1. Similarly, the FA0 cofactor is implemented using 2 input OR gate and sized the same way.

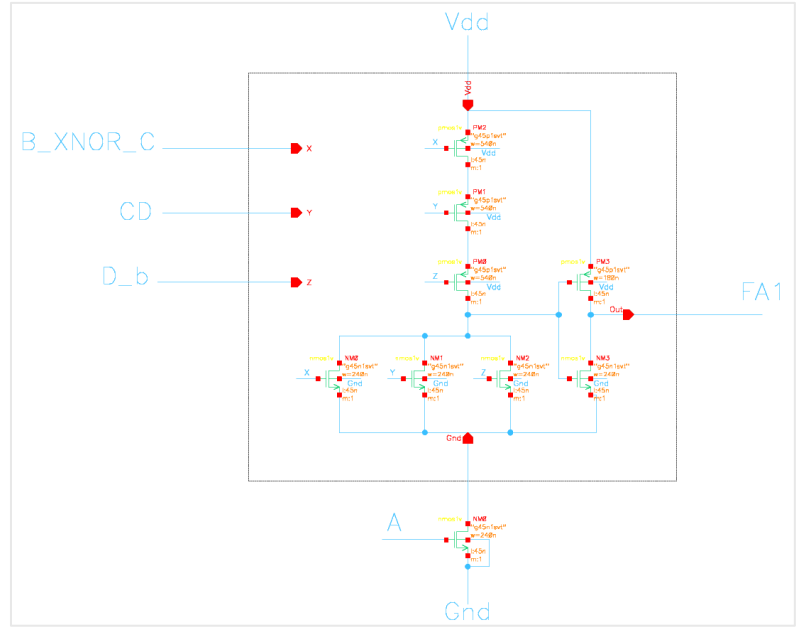


Fig 4: Sizing of pull up and pull-down for FA1

Leakage Current when all inputs are tied to '0' = 388.0 pA

Leakage Current when all inputs are tied to $V_{DD} = 220.6$ pA

Note: $V_{DD} = 1.1$ V

(e) Change the control variable to D and re-implement F using dynamic supply gating.

$$F_{D=1} = C + (A'B + AB') = C + (A \oplus B)$$

$$F_{D=0} = A + (B'C + BC') = A + (B \oplus C)$$

We observe that both the co-factors need same type of logic implementation (XOR followed by OR), hence we make that logic and size according to the reference inverter.

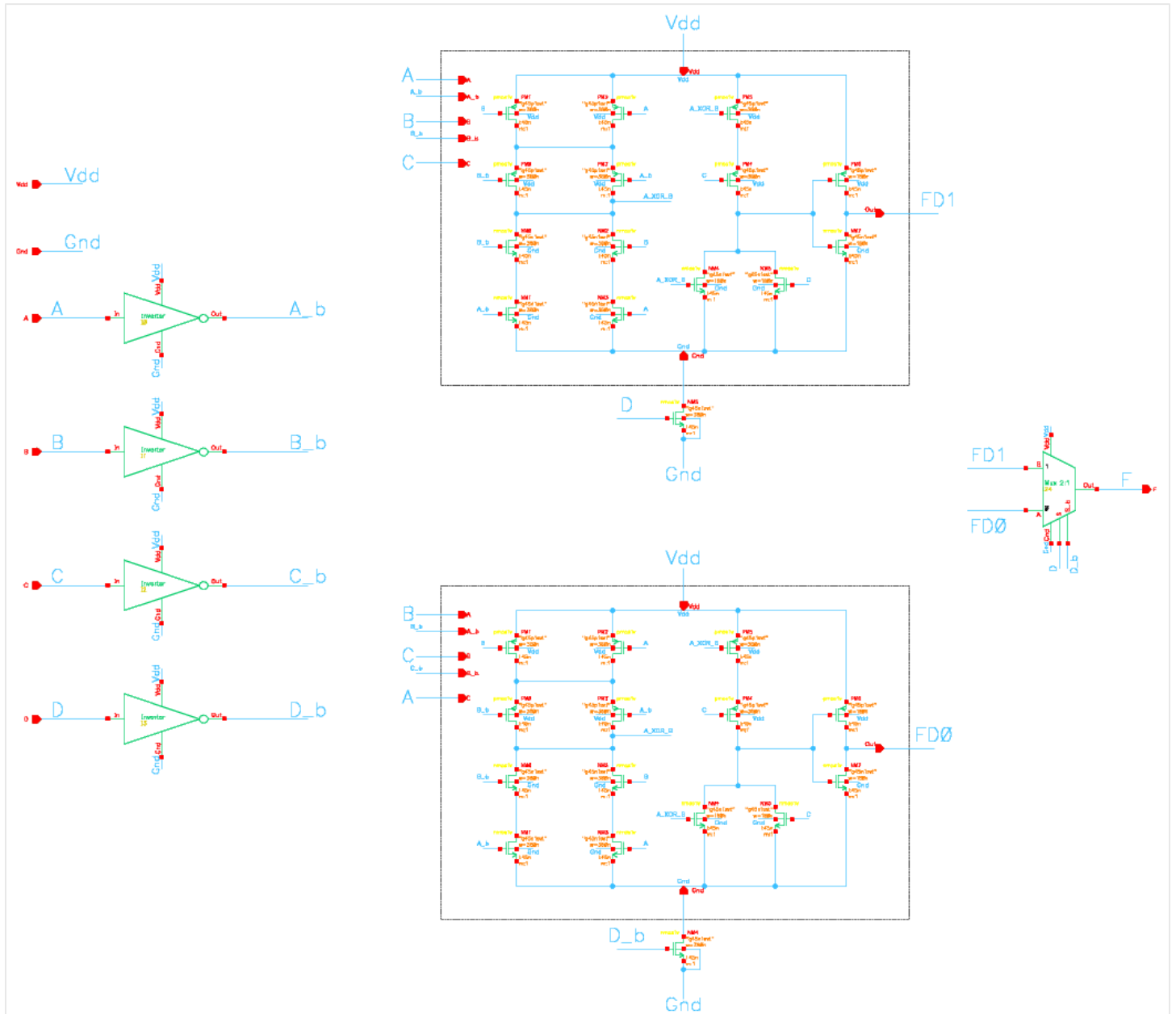


Fig 5: Function F with dynamic supply gating using variable D

(f) Size the transistors to balance the pull-up and pull-down networks and obtain the leakage when all inputs are tied to 0 and when all inputs are tied to VDD.

Firstly, Inverter was sized by keeping the width of NMOS $W_n = 120n$ and PMOS width W_p was swept to get equal rise and fall delays. We got $W_p = 180n$. Using this reference inverter, all other gates were sized using resistance equalization method.

Now, the pull-up and pull-down network for cofactor implementation (including the gating NMOS transistor) was sized according to resistance equalization to make the effective width of pull-up as $180n$ and pull-down as $120n$. This can be verified from Fig 6. which shows a XOR gate followed by OR gate for implementing FD1. Similarly, the FD0 cofactor is implemented using the same gate and same sizing.

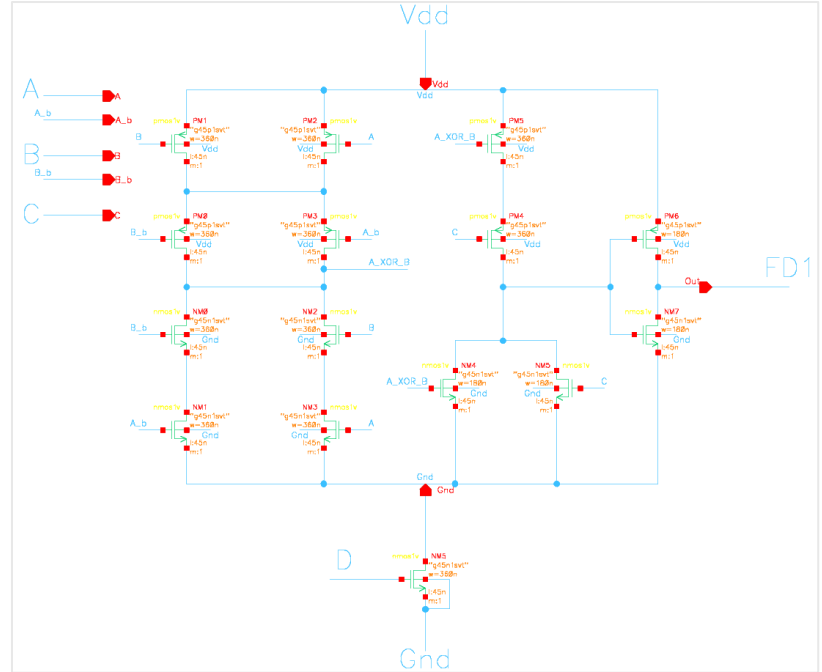


Fig 6: Sizing of pull up and pull-down for FD1/FD0

Leakage Current when all inputs are tied to '0' = 385.5 pA

Leakage Current when all inputs are tied to V_{DD} = 151.3 pA

Note: $V_{DD} = 1.1$ V

(g) Compare the leakage and the number of transistors for the standard implementation and the two implementations using dynamic supply gating.

Implementation	Number of Gates	Number of Transistors	Leakage (pA) when all inputs are 0	Leakage (pA) when all inputs are V_{DD}	Comments
Standard	5 NOT 2 NAND2 6 NAND3 1 NOR	29 PMOS 29 NMOS = 58	455.8	243.9	Baseline static CMOS implementation without supply gating. Highest transistor count and highest leakage in both all-0 and all-VDD input conditions due to all logic being always powered.
Dynamic Supply Gating with A variable	6 NOT 1 NAND2 1 XOR 1 OR2 1 OR3 1 MUX21 1 Gating	22 PMOS 23 NMOS = 45	388.0	220.6	Supply gating with A as the Shannon control reduces total transistor count and lowers leakage, especially when all inputs are 0. However, when all inputs are at VDD, the A=1 cofactor remains active, so leakage reduction is limited compared to the standard design.
Dynamic Supply Gating with D variable	4 NOT 2 XOR 2 OR2 1 MUX21 1 Gating	23 PMOS 24 NMOS = 47	385.5	151.3	Using D as the control variable provides better partitioning of the logic, resulting in significantly lower leakage when all inputs are at VDD. The transistor count is slightly higher than the A-gated case, but improved cofactor shutdown leads to better leakage savings for certain input vectors.